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Mock 2026 topics

1. Straight lines

Pythagoras' theorem and its converse.
Mid-point of a line segment and the distance between two points in the Cartesian plane.
The form $y = mx + b$ (gradient-intercept form).
Gradient; intercepts.
Lines with gradients $m_1 \; and \; m_2$ Parallel lines $m_1 = m_2$ and Perpendicular lines $m_1 \times m_2 = -1$
Perpendicular bisector.
Other forms of the equation of a straight line:
$ax + by + d = 0$ (general form).
$y - y_1 = m(x - x_1)$ (point-gradient form)

2. Quadratics

The quadratic formula. Solution of quadratic equations.
Graphs of quadratic functions. Vertex, axes intercepts, axis of symmetry.
Different forms of the quadratic function:
$f(x) = ax^2 + bx + c$ $\quad$ $f(x) = a(x - h)^2 + k$ $\quad$ $f(x) = a(x - x_1)(x - x_2)$
The discriminant $\Delta = b^2 - 4ac$ and the nature of the roots.
Quadratic inequalities.
Applications of quadratic functions.

3. Functions and Transformations

Functions, function notation, images. The graph of a function; its equation $y = f(x)$
Domain, range, intercepts, roots, maximum and minimum values.
Composite functions. Inverse function. Use of GDC.
Equations and graphs of a variety of functions.
The concept of a function as a mathematical model.
Using technology to graph functions including their sums and differences.
The graph of $y = f^{-1}(x)$ as the reflection in the line $y = x$ of the graph of $y = f(x)$.
Identity function.
Rational functions.
The reciprocal function $x \to \frac{1}{x}$, $x \neq 0$, its graph and self-inverse nature.
The rational function $x \to \frac{ax+b}{cx+d}$ and its graph.
Informal idea of limit. Limit notation. Vertical and horizontal asymptotes.
Polynomial functions: degree of a polynomial and equality of coefficients.
Equations and graphs of a variety of functions.
Finding the point of intersection of two curves or lines using technology.
Solving unfamiliar equations using technology.
Example: $2^x = x + 5$

4. Exponents and logarithms

Laws of exponents. Exponential equations.
Exponential functions and their graphs. Horizontal asymptote.
Informal idea of limit. Limit notation
Definition of logarithms. Laws of logarithms. Logarithmic functions and their graphs. Vertical asymptote. Exponential and logarithmic functions as one inverse of the other.
Logarithmic equations.
The number e. Natural logarithms.
Change of base.
Solving hidden quadratic equations with exponentials and logarithms.
Applications: Applications of graphing skills and solving
equations that relate to real-life situations.

5. Statistics : Basic Statistics and Correlation

Concepts of population, sample, random sample, discrete and continuous data.
Presentation of data: frequency distributions (tables); frequency histograms with equal class intervals.
Grouped data: use of mid-interval values for calculations; interval width; upper and lower interval boundaries; modal class.
Statistical measures and their interpretations.
Central tendency: mean, median, mode. Quartiles, percentiles. Dispersion: range, interquartile range.
Cumulative frequency; cumulative frequency graphs; use to find median, quartiles, percentiles.
Linear correlation of bivariate data.
Scatter diagrams; lines of best fit.
Equation of the regression line of y on x.
Use of the equation for prediction purposes. Pearson's product–moment correlation coefficient, r. Use of GDC.
Reliability of data sources and bias in sampling.
Sample techniques and their effectiveness: simple random, convenience, systematic, quota and stratified.
Measures of dispersion: standard deviation and variance.
Effect of constant changes to the original data.
Presentation of data including box and whisker plots and stem and leaf diagrams.
Outliers are defined as more than 1.5× IQR from the nearest quartile.
Types of correlation: Positive, zero, negative; strong, weak, no correlation. Linear and others.
Equation of the regression line of x on y.
Interpolation, extrapolation.
Mathematical and contextual interpretation.
Modelling a set of data points.
Solving a system of three equations with three unknowns.

6. Calculus 1: Basic differentiation

Introduction to the concept of a limit. Estimation of the value of a limit from a table or graph.
Derivative interpreted as gradient function and as rate of change.

Different forms of notation.
Informal understanding of the gradient of a curve as a limit.
Increasing and decreasing functions.
Graphical interpretation of the signs of the derivative function.
Identifying intervals on which functions are increasing or decreasing.
Derivative of expressions of the form x^n, n $\in$Z, polynomial functions, natural logarithm and exponential function.
Differentiation of a sum and a multiple of these functions.
Tangent and normal lines.
The second derivative. Higher derivatives.
Maximum and minimum points. Points of inflexion with zero and non-zero gradients.
Graphical behaviour of f, f', f''.
Use of the terms "concave-up" for f ″(x) > 0, and "concave-down" for f″ (x) < 0.
Function analysis.

7. Geometry and trigonometry I and II

Right-angled trigonometry. Angles of elevation and depression.
The distance between two points in three-dimensional space, and their midpoint.
Volume and surface area of three-dimensional solids including right-pyramid, right cone, sphere, hemisphere and combinations of these solids.
The size of an angle between two intersecting lines or between a line and a plane.
The sine rule. The cosine rule. Area of a triangle. The ambiguous case.

The unit circle. Radians measure. Length of an arc; area of a sector. Definition of sine and cosine in the unit circle. Definition of tangent as $\frac{\sin x}{\cos x}$
The Pythagorean identity $\sin^2 x + \cos^2 x = 1$
Exact values of trigonometric ratios of 0, $\frac{\pi}{6}$, $\frac{\pi}{4}$, $\frac{\pi}{3}$, $\frac{\pi}{2}$, and their multiples
Simple trigonometric equations with and without a calculator.
Trigonometric functions: their domains and ranges; amplitude, period. Their graphs.
Transformations of trigonometric functions.
Modelling using trigonometric functions.